# 1.2 Parent Functions and Transformations

Name _____ Period ____ Date _____

Algebra II 1.2 · Student sheet

**I can…**

- identify the family of a function.
- describe translations, reflections, stretches, and shrinks.
- describe combinations of transformations and find the domain and range.

**Vocabulary** · fill in the missing word

**Parent function** — the _____ function in a family (constant, linear, absolute value, quadratic, …)

**Transformation** — changes the size, shape, position, or _____ of a graph

**Translation** — _____ a graph left, right, up, or down

**Reflection** — _____ a graph over a line

**Vertical stretch / shrink** — _____ every y-value by a > 1 / by 0 < a < 1

## Part 1 · Parent function gallery

Graph $y = |x|$ in Desmos. Then graph each function and compare it with $y = |x|$.

| Function | Vertex | What changed? |
|---|---|---|
| $y = \|x\| + 2$ | | |
| $y = \|x - 3\|$ | | |
| $y = -\|x\|$ | | |
| $y = 2\|x\|$ | | |
| $y = \frac{1}{2}\|x\|$ | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

### Example 1

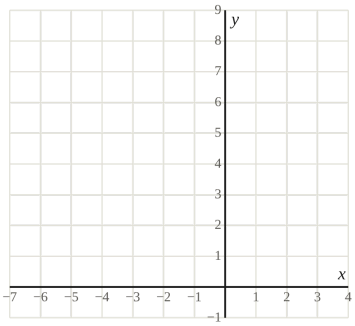
Identify the function family of each function.

a. $f(x) = |x - 1|$ b. $g(x) = -3x^2 + 5$ c. $h(x) = 2x - 7$ d. $k(x) = 6$

**Translations:** $f(x) + k$ moves the graph _____; $f(x - h)$ moves it _____.

### Example 2

Describe the transformation of $f(x) = x^2$ represented by $g(x) = (x + 3)^2$. Then graph both.

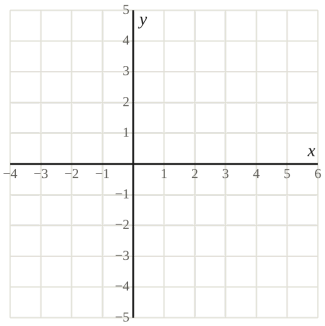



**Reflections:** $-f(x)$ is a reflection in the _____; $f(-x)$ is a reflection in the _____.

**Stretches and shrinks:** $a \cdot f(x)$ with $a > 1$ is a vertical _____; with $0 < a < 1$ it is a vertical _____.

### Example 3

Describe the transformation of $f(x) = |x|$ represented by $g(x) = -2|x - 1| + 3$.




### Example 4

Give the function family, domain, and range of $f(x) = (x - 2)^2 - 5$.

## Part 3 · Partner practice

### Partner 1

Describe the transformation of $f(x) = x^2$ represented by $g(x) = \frac{1}{2}x^2$.

### Partner 2

Describe the transformation of $f(x) = |x|$ represented by $g(x) = -|x + 4| - 2$. Give the range.